

	STEPS	REASON
d)	$a \wedge b$ $\neg x$ $a \rightarrow (r \wedge b)$ $(r \vee n) \rightarrow (x \vee p)$ $\therefore p$	

7. Use contradiction to prove the following argument:

	STEPS	REASON
$\left\{ \begin{array}{l} p \rightarrow q \\ (q \wedge r) \rightarrow s \end{array} \right.$ $\textcircled{f}$ $\therefore p \rightarrow s$	1) $\neg (p \rightarrow s)$ 2) $\neg (\neg p \vee s)$ 3) $\neg \neg p \wedge \neg s$ 4) $p \wedge \neg s$ 5) p 6) $\neg s$ 7) $p \rightarrow q$ 8) q 9) r 10) $q \wedge r$ 11) $(q \wedge r) \rightarrow s$ 12) s 13) $p \rightarrow s$	negating conclusion, doing proof by contradiction no NAME 1 DeMorgan's 2 Double neg. 3 C.S. 4 C.S. 4 given M.P. 5, 7 given R.C. 8, 9 given M.P. 10, 11 Lines 6, 12 are in contradiction.

If all of the problems can't be done in the lab period, students can work on them on their own.

	STEPS	REASON
d)		
$a \wedge b$	1) $\neg p$	negating conclusion
$\neg x$		prog by contradiction
$a \rightarrow (r \wedge b)$	2) $(r \vee n) \rightarrow (x \vee p)$	given
$(r \vee n) \rightarrow (x \vee p)$		
$\therefore p$	3) $\neg x$	given
	4) $\neg p \wedge \neg x$	P.C. 1,3
	5) $\neg (p \vee x)$	De Morg- 4
	6) $\neg (r \vee n)$	M.T. 2,5
	7) $\neg r \wedge \neg n$	De Morg. 6
	8) $\neg r$	C.S. 7
	9) $\neg n$	C.S. 7
	10) $a \wedge b$	given
	11) a	C.S. 10
	12) b	C.S. 10
	13) $a \rightarrow (r \wedge b)$	given
	14) $r \wedge b$	M.P. 11,13
	15) r	C.S. 14
	16) p	lines 8, 15 are in contradiction